

2022.

Q5.

$$(a). |H(j\omega T)| = \left| \frac{10 \cdot j\omega T + 1}{20 \cdot j\omega T + 1} \right| = 0.501 = -6.0 \text{ dB.}$$

$$\omega_N = \frac{2\pi}{T} \times \frac{1}{2} = 0.28 \text{ rad/s} \quad f_N = 1 \text{ Hz.}$$

$$(b). H(s) = \frac{Y(s)}{U(s)} = \frac{10s+1}{20s+1} \Rightarrow 20s Y(s) + Y(s) = 10s U(s) + U(s)$$

$$\Rightarrow 20 \dot{y}(t) + y(t) = 10 \dot{u}(t) + u(t)$$

$$\Rightarrow 20 \cdot \frac{y_k - y_{k-1}}{T} + y_k = 10 \cdot \frac{u_k - u_{k-1}}{T} + u_k$$

$$\Rightarrow \frac{20}{T} (Y(z) - Y(z) \cdot z^{-1}) + Y(z) = \frac{10}{T} (U(z) \cdot z^{-1} - U(z) \cdot z^{-1}) + U(z)$$

$$\Rightarrow \frac{Y(z)}{U(z)} = \frac{\frac{10}{T} (1 - z^{-1}) + 1}{\frac{20}{T} (1 - z^{-1}) + 1} + U(z)$$

$$= \frac{21 - 20 z^{-1}}{41 - 40 z^{-1}}$$

$$= \frac{21z - 20}{41z - 40}$$

$$= 0.5122 \cdot \frac{z - 0.9524}{z - 0.9756}$$

(c).

$$H(z) = H(s) \left. \frac{\omega_d}{\omega_n} \right|_{s = \frac{2}{T} \frac{z-1}{z+1}} = H(s') \left. \frac{\omega_d}{\omega_n} \right|_{s' = \frac{\omega_d}{\omega_n} \cdot \frac{2}{T} \frac{z-1}{z+1}}$$

$$= H(s') \left. \frac{\omega_d}{\frac{2}{T} \tan \frac{\omega_d T}{2}} \cdot \frac{2}{T} \cdot \frac{z-1}{z+1} \right|_{s' = \frac{\omega_d}{\tan \frac{\omega_d T}{2}} \cdot \frac{z-1}{z+1}}$$

$$H(s) = \frac{10 \cdot \tan^{-1} \frac{\omega}{2} \cdot \frac{z-1}{z+1} + 1}{20 \cdot \frac{\omega_d}{\tan \frac{\omega_d}{2}} \cdot \frac{z-1}{z+1} + 1} = \frac{10 \times 3.22 \times \frac{z-1}{z+1} + 1}{20 \times 3.22 \cdot \frac{z-1}{z+1} + 1}$$

$$= 0.5076 \frac{z - 0.9599}{z - 0.9694}$$

7.022

Q1. (b). design goals:

- ① Close loop stability : Nyquist plot of L_0 doesn't encircle -1 .
- ② stability Robustness : make S_0 and T_0 small in different frequency ranges.
- ③ Disturbance Attenuation: Avoid peaking of S_0
- ④ Command Response : T_0 , F , command tracking.
- ⑤ Plant capacity : u cannot be too large.
- ⑥ Measurement Noise : Avoid peaking of T_0
- ⑦ Performance robustness.

Limit :

- ① plant capacity,
- ② measurement noise.

(c).
$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -0.3 & -4.7 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

$$= \begin{bmatrix} -0.3 X_1(t) - 14.7 X_2(t) + u(t) \\ X_1(t) \end{bmatrix}$$

$$\Rightarrow \begin{cases} \dot{X}_1(t) = -0.3 X_1(t) - 14.7 X_2(t) + u(t) \\ \dot{X}_2(t) = X_1(t) \end{cases}$$

$$\Rightarrow \ddot{X}_2 = -0.3 \dot{X}_2(t) - 14.7 X_2(t) + u(t)$$

$$\Rightarrow s^2 X(s) = -0.3 s X(s) - 14.7 X(s) + U(s)$$

$$\Rightarrow \frac{X(s)}{U(s)} = \frac{1}{s^2 + 0.3s + 14.7}$$

$$y(t) = \begin{bmatrix} 0 & 0.6 \end{bmatrix} \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} = 0.6 X_2(t)$$

$$\Rightarrow Y(s) = 0.6 X(s) = \frac{0.6}{s^2 + 0.3s + 14.7}$$

$$\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow x_2(0) = 1$$

The initial states

(d). Controllability :

a state x is said to be controllable if for $x_0 \neq 0$ and for any finite time interval $[0, T]$ there exists an input $\{u(t), t \in [0, T]\}$,

such that $x(T) = 0$,

If all states are controllable,

then the system is said to be completely controllable.

The test of controllability

$$\text{Given } \begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

the controllability matrix

$$T_c(A, B) = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

has full row rank (i.e. non-singular).

it is C.C.

Q3.

$$\text{ex). } p(j\omega) = \frac{0.1}{(1 + \frac{j\omega}{10})(1 + \frac{j\omega}{100})}$$

$$\omega_1 = 10 \text{ rad/s} \quad \omega_2 = 100 \text{ rad/s.}$$

$$20 \log_{10} [p(j\omega)] = 20 \log 0.1 - 20 \log \left(1 + \frac{j\omega}{10}\right) - 20 \log \left(1 + \frac{j\omega}{100}\right)$$

$$-20 \text{ dB} \quad \omega < \omega_1$$

$$(-20 - 20\omega) \text{ dB} \quad \omega_1 \sim \omega_2$$

$$(-20 - 40\omega) \text{ dB} \quad \omega > \omega_2$$

$$\angle p(j\omega) = -\arctan \frac{\omega}{T_D} - \arctan \frac{\omega}{T_2}$$

$$(b), \quad u(t) = K \left(e(t) + \frac{1}{T_2} \int_0^t e(\tau) d\tau + T_D \dot{e}(t) \right)$$

$$\Rightarrow U(s) = K \left(E(s) + \frac{1}{T_2} \cdot \frac{1}{s} E(s) + T_D s E(s) \right)$$

$$\Rightarrow C(s) = \frac{U(s)}{E(s)} = K \left(1 + \frac{1}{T_2 s} + T_D s \right)$$

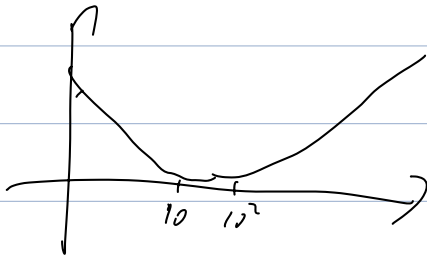
$$= \frac{K}{s} \left(s + \frac{1}{T_2} + T_D s^2 \right)$$

$$\approx \frac{K}{s} \left(T_D s^2 + \left(1 + \frac{T_D}{T_2} \right) s + \frac{1}{T_2} \right)$$

$$= \frac{K}{s} (T_D s + 1) \left(s + \frac{1}{T_2} \right)$$

$$\omega_1 = \frac{1}{T_D} = 100 \text{ rad/s}$$

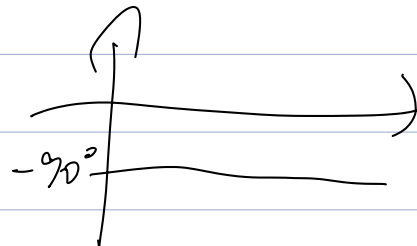
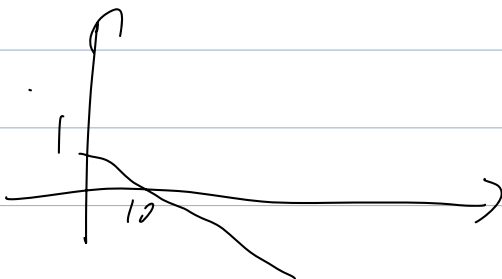
$$\omega_2 = \frac{1}{T_2} = 10 \text{ rad/s}$$



$$C(s) = \frac{10 (0.01s + 1) (s + 10)}{s}$$

$$= \frac{0.1 (s + 100) (s + 10)}{s}$$

$$L(s) = C(s) \cdot P(s) = \frac{10}{s}$$



$$Q6. (a). G_{ZAS}(z) = (1 - z^{-1}) Z\left(\frac{G(s)}{s}\right)$$

$$= (1 - z^{-1}) Z\left(\frac{1}{s(0.1s+1)}\right)$$

$$Z\left(\frac{G(s)}{s}\right) = Z\left(\frac{A}{s} + \frac{B}{0.1s+1}\right)$$

$$A(0.1s+1) + Bs$$

$$= (0.1A+B)s + A = 1$$

$$\Rightarrow A = 1$$

$$B = -0.1$$

$$= Z\left(\frac{1}{s} - \frac{1}{s+10}\right)$$

$$= \frac{z}{z-1} - \frac{z}{z-e^{-1}}$$

$$= 1 - \frac{z-1}{z-e^{-1}} = \frac{1-e^{-1}}{z-e^{-1}}$$

$$= \frac{0.6321}{z-0.3679}$$

(P) PI controller

$$C_{PI}(z) = K_P + K_I \cdot \frac{z}{z-1} = \frac{K_P(z-1) + K_I \cdot z}{z-1}$$

$$\frac{C_{PI}(z) \cdot G_{SM1}(z)}{1 + C_{PI}(z) \cdot G_{SM1}(z)} = \frac{(K_P + K_I)z - K_P}{z-1}$$

$$= \frac{0.6321[(K_P + K_I)z - K_P]}{(z-1)(z-0.3679) + \overset{0.6321}{(K_P + K_I)z} - \overset{0.6321}{K_P}}$$

$$= \frac{\text{~~~~~}}{\text{~~~~~}}$$

$$z^2 + [0.6321(K_P + K_I) - 1.3679]z$$

$$+ 0.3679 - 0.6321K_P$$

$$-2\cos(\omega_d T) \cdot e^{-\zeta \omega_n T} = 0.6321(K_P + K_I) - 1.3679$$

$$e^{-2\zeta \omega_n T} = 0.3679 - 0.6321K_P$$

$$\Rightarrow K_P = 0.3679$$

$$K_I = 1.9830$$